Cob-crete Molds

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Abstract:

This project looks at the development of reusable concrete and cob molds for specimen fabrication. Existing molds are single use, create unnecessary waste, and can be challenging to remove. With the current mold, pressurized air is often used for removal, if the pressurized air is unable to remove the concrete specimen, the whole mold would need to be cut open. Likewise, with cob, the mold would be needed to be cut open every time. There were many approaches that we took in this project, push-up molds, 3D molds, wood and cardboard molds. Testing focused on ease of removal, functionality, how water tight molds were based on need of the specimen, and if the molds could be reused again. Results showed that the reusable molds have high potential as alternatives to regular single use molds. However, there needs to be more tests done to show that the molds along with liners could be used long term.

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Problem:

The problem our client and partner, Professor Hana Mori, presented to us was the need for reusable molds for both cob and concrete fabrication processes. The molds she currently uses for these materials are single-use, making them wasteful. In addition, the existing molds make it very difficult to remove finished specimens without damaging the molds themselves.

For example, when working with concrete, pressurized air had to be used to release the specimen from the mold. However, without proper and consistent technique, the mold could easily crack, explode, or break during the process. For cob specimens, the mold must be cut off entirely in order to remove the material, which results in unnecessary waste.

Design Goals:

What is the need/lack/problem as described by your client?

1. Specimen quality: The molds must comply with the requirements outlined in ASTM C470.
    a. Concrete mold dimensions: 4 inches in diameter and 8 inches in height
    b. Cob mold dimensions: 6 inches in diameter and 12 inches in height.
    c. Top and bottom surfaces of the molds must be flat and parallel, with sides that are perpendicular to ensure accurate specimen geometry.
    d. The finished specimens must also have smooth surfaces with minimal imperfections, such as ridges or bumps, since these irregularities can compromise the accuracy of compressive strength testing.
2. Reusability and waste reduction
    a. Current molds are single-use, meaning they must be discarded after each specimen is produced. This results in unnecessary material waste and limits the efficiency of the testing process.
3. Ease of removal and reuse
    a. The current process for removing concrete and cob specimens is inefficient and often damages the mold. To address this issue, our design should allow users to remove specimens quickly and safely without damaging the mold nor specimen, therefore making the process more efficient and practical for repeated use.
4. Accessibility
    a. Create a reusable mold from materials that are affordable and available. Many sustainable concrete molds that are currently on the market require a large

downpayment or heavy machinery to create a similar design. We must make something easy and accessible to the general public.

We decided to split our design into two different materials: cob and concrete. As mentioned previously, while materially they are similar, they both have different processes to complete them. Cob needs to lose moisture during its drying process, while concrete must maintain moisture throughout.

COB

Proposed Solution:

For the cob molds, our initial and main concept for a reusable design was to incorporate a push-up mechanism within the mold. The idea was to place a circular disc inside the mold, with a diameter slightly smaller than that of the cylindrical cavity. It would be locked in place due to a lip at the bottom of the cylinder mold. This disc would be able to slide vertically within the mold,allowing the user to push the cob specimen upward and out of the mold—similar to the mechanism of a push-pop container.

To support this design, the mold would also include removable liners to help the cob maintain its shape during fabrication and curing. These liners would reduce friction between the material and the mold walls, making it easier to release the specimen while also preserving its structural integrity during the forming process.

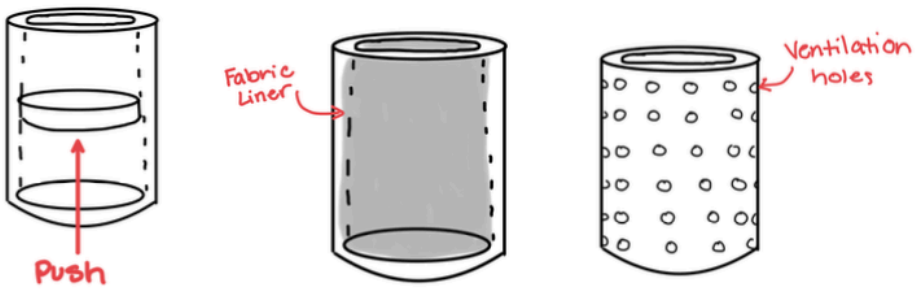


Figure 1: Concept sketch of the cob push-up mold mechanism designed to allow specimen removal without damaging the mold.

Design Process:

For reusability and feasibility we focused on:

1. Standardizing sizing for all samples to 3”x6” for initial samples and later 6”x12”
2. Using cleanable products such as fabrics and plastics to minimize long term waste
3. Making simple user friendly designs so the switch is as clean as possible

For our testing process, we decided to use three different sets of fabrics that could be potentially used for the removable, reusable liners. We used flannel, fleece, silk, satin, cotton, and felt. Silk, cotton, and satin were sewn together, while flannel and felt were by themselves. To test which fabrics were good at extracting moisture from the cob specimen, we created three cylinders and placed one of each liner set inside of them.



Figure 2: *Fabric materials used for the cob mold liners. From Left to Right we have a Cotton Satin mix, Cotton fabric, and a Fleece and Towel Mix.*

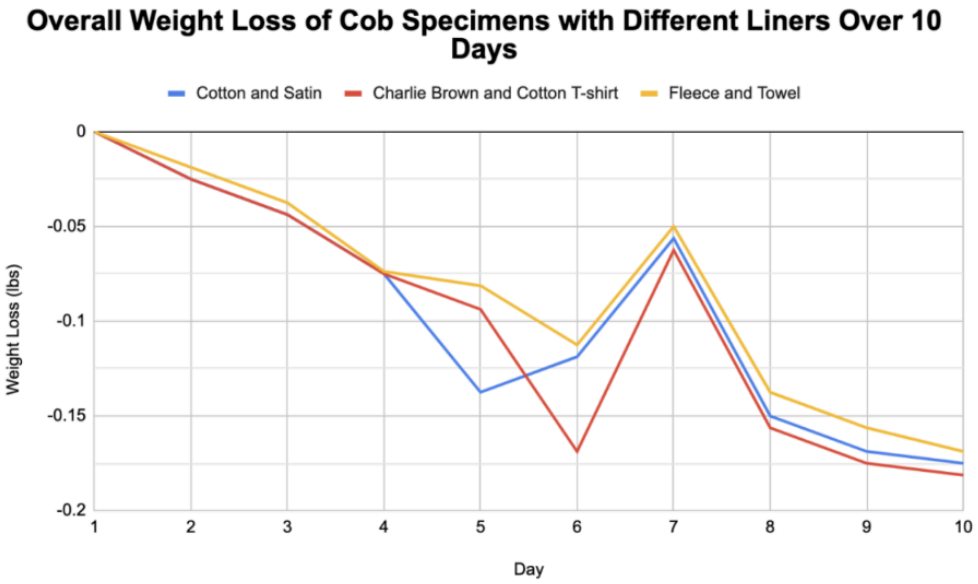


Figure 3: *Weight Loss of COB specimens over a 10-day drying period using three different liner material combinations (shown in Figure 1).*

We tested each of the molds with the mold design shown in Figure 1 over a ten day span. The weight changes in figure 2 reflect the moisture loss of each specimen allowing us to see how the liners may slow down or accelerate the drying process. Since they were dried outside, we can see a large spike in our dataset from a sharp increase in humidity through days 6-8.

Our second test was focused on the mold design rather than the liner material. We used the cotton fabric material for all of our tests as it showed the most moisture loss over the 10 day testing period.

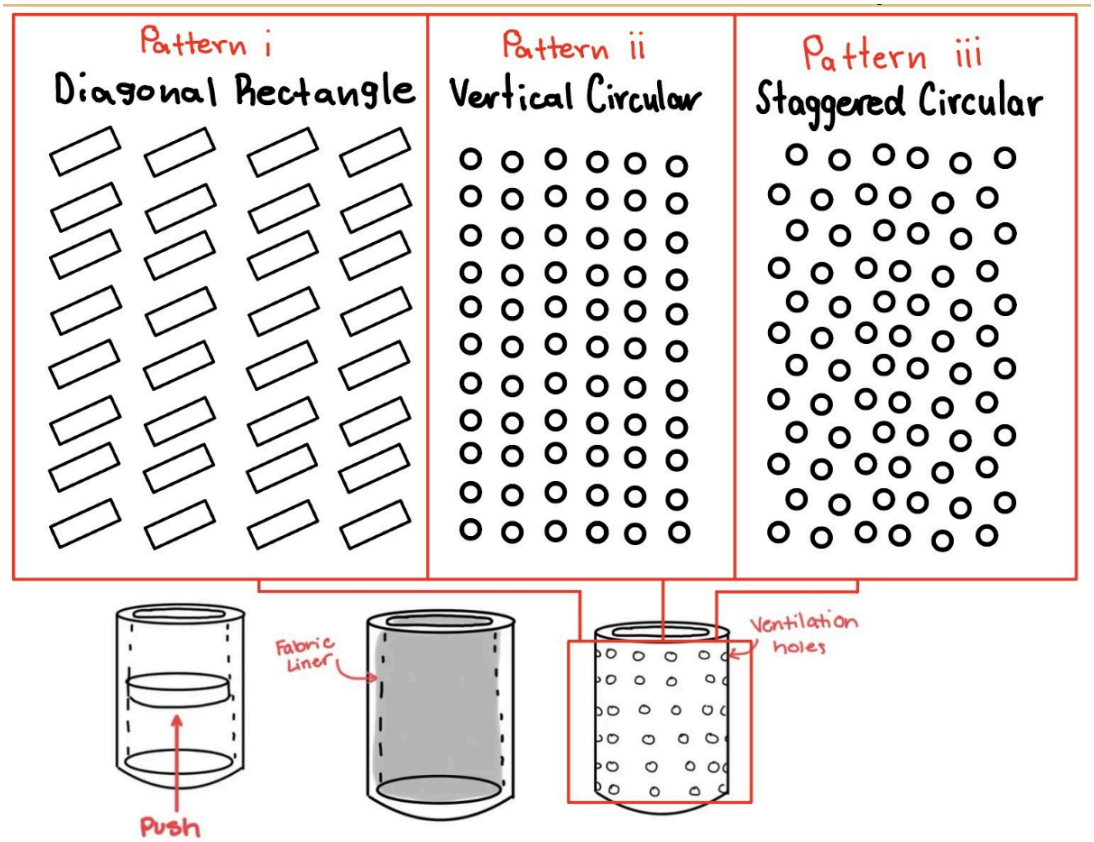


Figure 4: *Shows our 3 ventilation patterns used to test how the mold design affects moisture release from cob specimens*

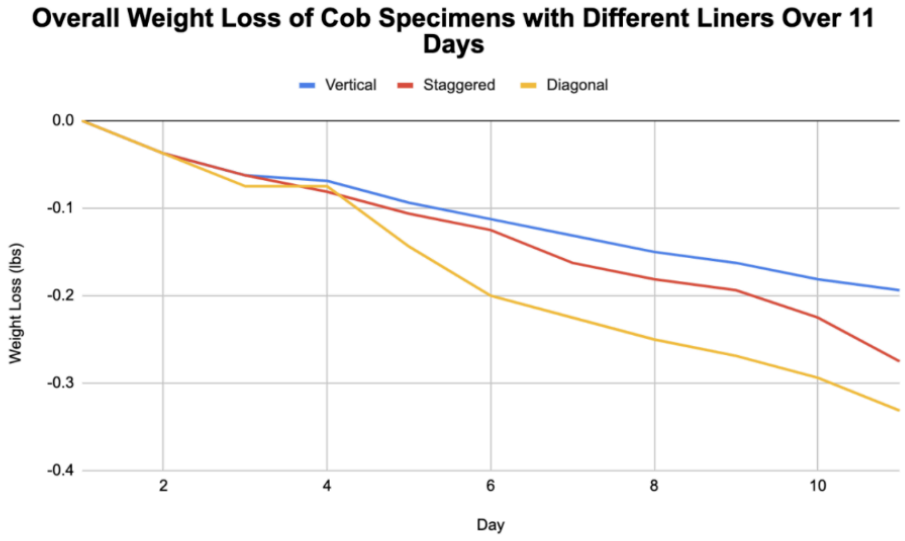


Figure 5: *Weight loss of COB specimens over an 11-day drying period using our 3 different ventilation combinations*

The COB specimen with the greatest weight loss while preserving the shape and consistency of the specimen was the combination of the felt liner and the push-up mold with diagonal rectangular holes. The final fabrication and assembly process is shown below.

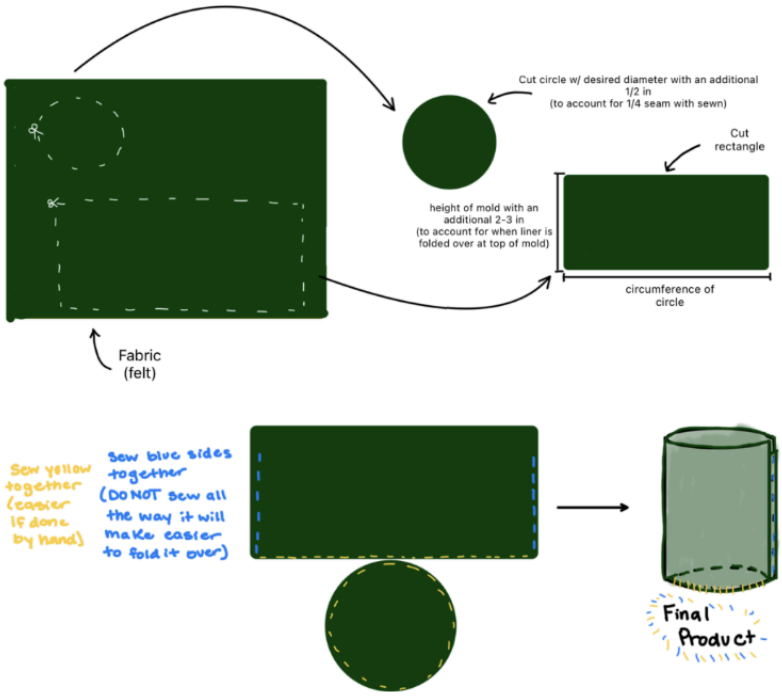


Figure 6: *Diagram showing the fabrication process for the cylindrical fabric liner*

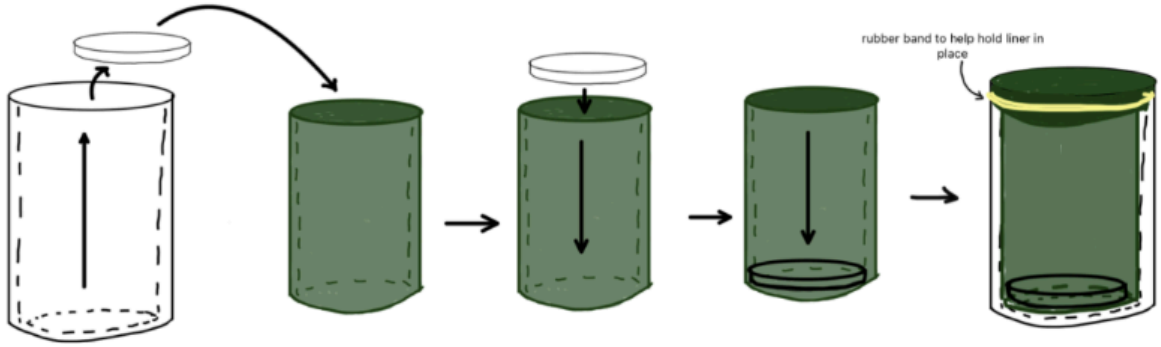


Figure 7: *Assembly process for the cob mold system showing the placement of the liner held in place by a rubber band during the pouring process.*

Concrete

While there are existing reusable designs they are often expensive, difficult to use, and limited to specific measurements. Our goal was to create sustainable and accessible molds for the general public. To investigate sustainable and reusable alternatives to traditional concrete formwork systems, several prototype mold configurations were designed and tested using additive manufacturing, clay-based liners, cardboard systems, and wood-supported assemblies. The development process was informed by existing research on reusable and low-waste formwork systems, particularly studies involving clay-based formwork, reusable casting systems, and adaptable fabrication methods for concrete elements.

Preliminary Research and Design Goals:

Our research was focused in two sections: for adaptable liners that can be added to preexisting systems, and for new systems that utilize recycled materials for a reusable mold product. Unlike the COB molds, concrete brought significant difficulties and limitations in materials we may use and design restrictions in order to fully preserve the specimen strength.

The molds for specimens in contact with concrete are generally made of steel, cast iron, or other nonabsorbent materials, nonreactive with concrete. They must hold their dimensions and shape under all conditions of use and be watertight–judged by their ability to hold water poured into them for approval. In some cases sealants such as grease, modeling clay, or microcrystalline wax may be used and generally reusable molds are lightly coated in mineral oil or another nonreactive form. Other materials such as paper liners are also prohibited for standard-cured specimens.

Given these limitations we initially focused our research towards lining materials for existing molds. This involved looking into materials such as various mineral oils, polypropylene, acrylonitrile butadiene styrene (abs), metal sheets, candle wax, HDPE sheets, and dissolvable liners such as PVA film.

Further research brought us the opportunity to look into more modern research with architectural molds using recycled materials such as a resin coated cardboard formed molds made entirely of type 2 wood glue, sawdust, and recycled cardboard. The projects were initially used for projects such as architectural pieces and home goods however due to the scope of this project, making a singular mold required further time than we had available.

Another example of modern architectural research was to use recycled clay and wood formwork to create concrete molds. These projects used clay 3D printing to create intricate designs that general molds were unable to, allow placement of wooden formwork to support the structure, and pour concrete allowing the greater shrinkage rate of clay to simply fall off the concrete as the samples naturally dry–completely eliminating a “demolding” process and allowing the used to regather and rehydrate to clay for another project.

Our goals were to test both aspects of modern architectural designs as well as easily applied lining materials for our designs to contribute to both the accessible materials and ease of use for our client.

Design Process:

3D Printed Designs

We began the design process by creating a general mold design focused on water tightness and ease of use. To create a simple, adaptable design, we used 3D-printed molds with a split cylinder, a removable bottom, and a fastened lid, all held together by two adjustable hose clamps for uniform compression.

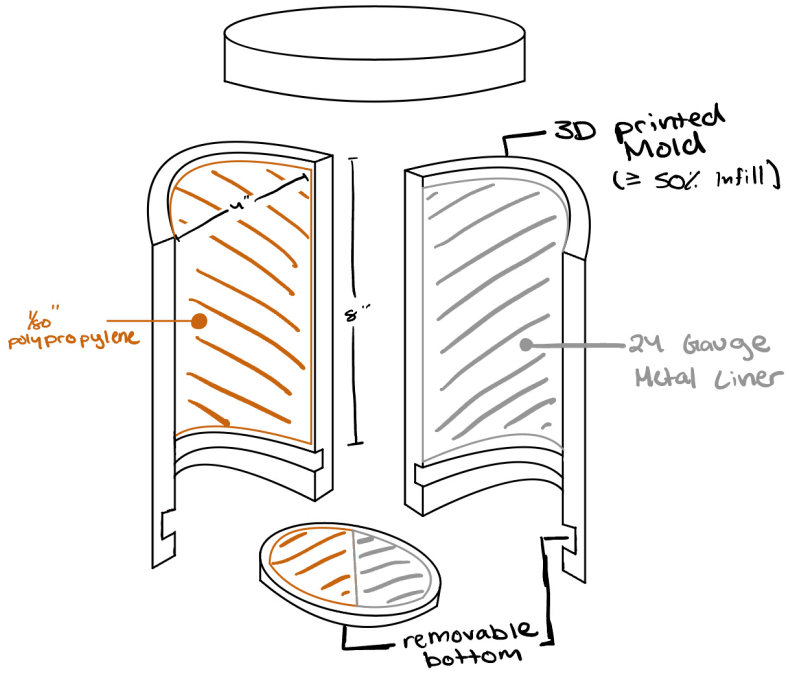


Figure 8: *Showing the setup of our 3D printed molds with liner placements*

The snap-on lid was a customizable piece that provided an airtight seal to the top of the mold while improving the compression towards the top. This helped improve watertightness (as seen in later testing) and gave room for our additional liners. It was easily removable and fit for all internal liners and materials that we added.

We began testing this initial design to evaluate whether our mold design reduces leakage and improves sealing performance compared to existing molds.

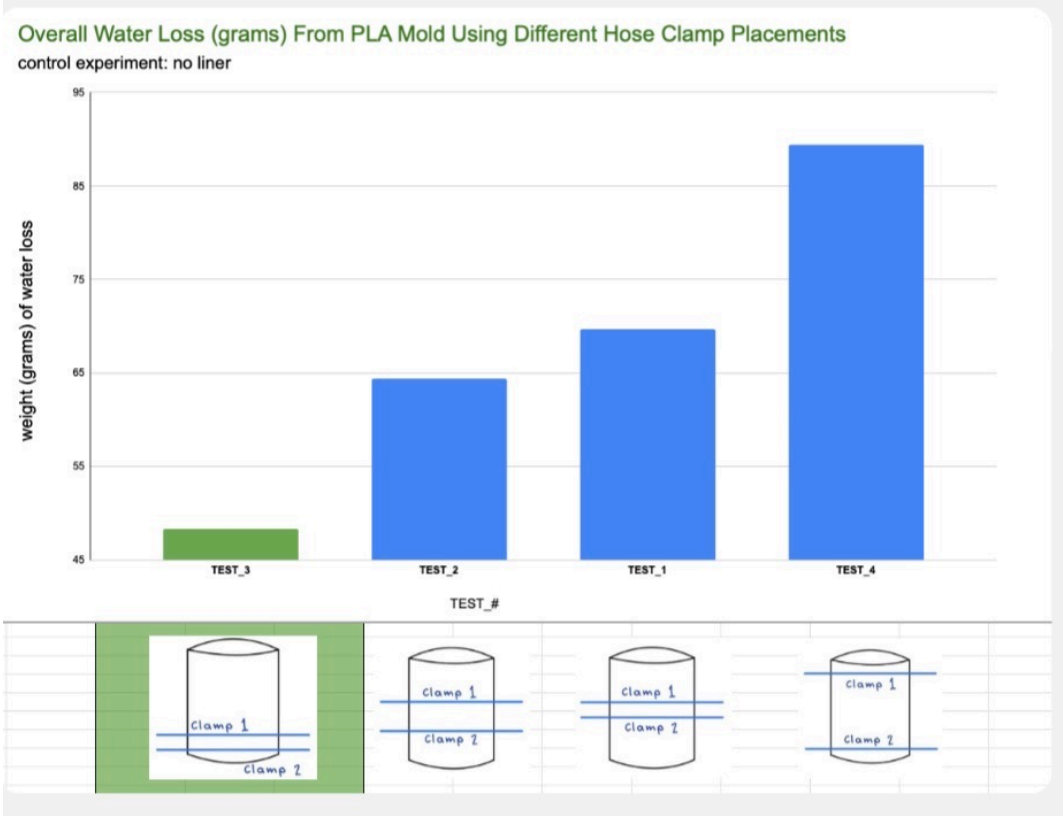


Figure 9: *Shows the water leakage over 1 minute using 500 grams of water.*

Based on the water leakage data from figure #, we chose to use placement # with one hose clamp over the removable bottom and the second two inches above that point. We also added silicone sealant along all the open edges.

For our lining materials we chose polypropylene (PP), acrylonitrile butadiene styrene (ABS), 24 gauge aluminum sheet metal, 26 gauge sheet metal, and a thin plastic wrap like liner. However, the thinner plastic was not used for final testing since it required a separate mold and attachment process that was too intensive to qualify for our third design goal “Ease of Reusability and Reuse”. We came across a similar issue with any sheet metal 26 gauge and above since it was too thick to be easily molded to the dimensions we needed without heavy equipment, therefore

breaking our fourth design goal “Accessibility”. In the end, we choose the PP plastic, ABS, and the 24 gauge aluminium sheet metal as liners.

Our final three liners along with the silicone lining were then tested again for water leakage under the same conditions as figure 9.

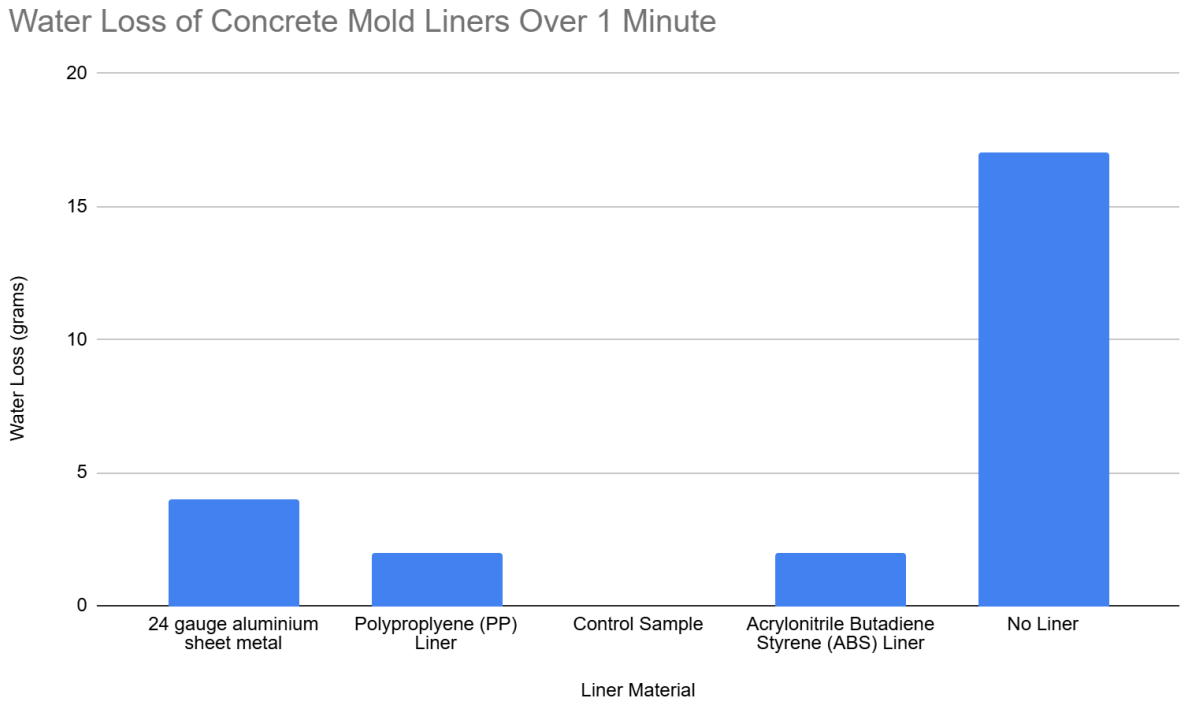


Figure 10: *This graph shows the water loss out of 500 grams of water over 1 minute for our 3D printed molds with the silicone sealant and our chosen liners*

Clay Formwork

This wood-supported mold used a hybrid fabrication combining the structural support from the wooden framework to the weaker clay liners.



Figure 11: *Wood mold with clay inside for concrete to be poured into.*

The product worked by creating a wooden framework just over the ASCE standard at 6.5”x13” leaving room for the clay. Once cut, glued and stained, the clay was added and shaped to fit the clay with 6”x12” of room. This meant 3D printing a press for the concrete sample shape and filling in the gaps with low fire ceramic clay. Once the clay is left out to dry for 1-3 days, the concrete can be poured. For the first 24-36 hours the wooden framework is attached to the sample.



Figure 12: Shows how the clay is breaking off as shrinkage happens.

After this point the wooden framework can be taken out leaving only the clay and concrete materials. Since the clay has a faster shrinkage rate than the concrete. It will continue to naturally fall off during the rest of the drying process.



Figure 13: *shows the final sample surrounded by the dried and peeling clay*

The wooden framework can then be cleaned and reused and the clay can be rehydrated for the next sample.

Concrete Pouring

Goal*: Evaluate how well will each of the molds be when used to make concrete? We will be looking at ease of use, how uniform each specimen will turn out and if there are any residues left behind.*

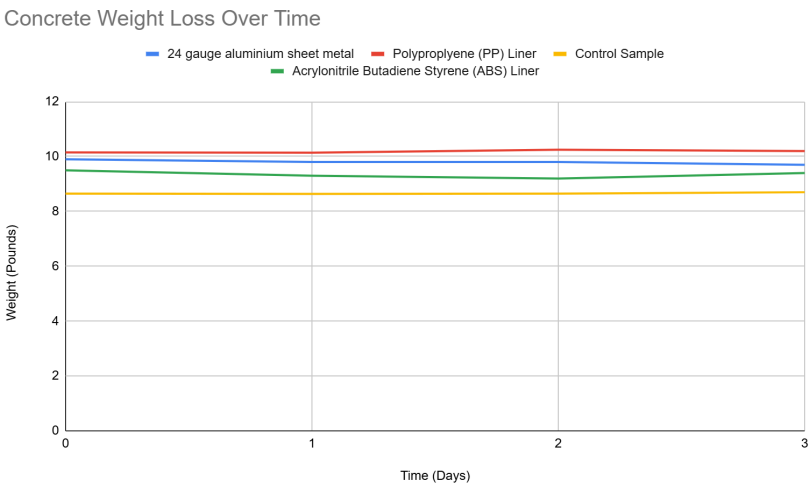
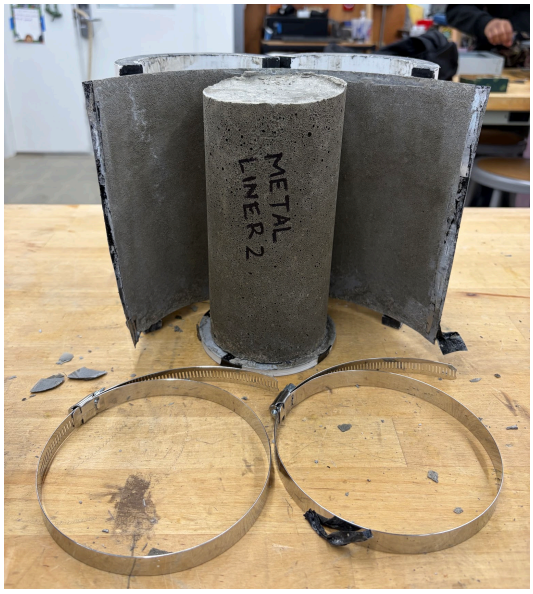


Figure 14: *Graph of weight loss over time*

We poured in concrete and let it cure for a maximum of three days across all types of liners and molds to see how easy it would be to take out and the residue left behind.

Throughout all of the molds, there was very little water loss, this was expected as previous tests showed that water loss will not be an issue. To determine which liner would be used in the Final Design, we will be looking at qualitative data.

Liner	Observations	Sample Photos
24 Gauge Aluminium Sheet metal	It was easy to take out the specimen. There was a lot of residue left over from the concrete pours, and it was not able to be wiped off. We had to be careful to not crease the sheet metal.	

Polypropylene Plastic	The flexible plastic allowed us to take out the concrete very easily, we did not have to worry about warping the liner. There was very little residue left over, we were able to wipe what was left very easily.	
Control	The control group was the hardest to take out. With the two tests done, we were able to only successfully take out the concrete once without cutting open the mold.	
ABS Plastic	The concrete was very easy to remove, however, there was still residue leftover and it was hard to remove as compared to the Polypropylene liner.	

Over the two cycles of three day pouring periods, in between each day, we weighed out the concrete to check if there was water loss in the weight of each specimen. The goal was to maintain the same amount of water weight across the three days. At the end of the curing process, we took out each of the specimens from their molds and checked on how easy each mold was to take apart, how the concrete turned out, and if the liners were in good condition. From this test, we determined that with all the liners including the control group, the concrete all had a smooth surface with very little imperfections. The main difference between all the tests was how the liners turned out. First, with the metal liner, we were only able to do two tests max before having to replace it. The concrete would bind to the liner and it was hard to wipe off. Next, the PLA plastic liner did better, there was less concrete remaining on the liner, but there was some left still. Finally, the polypropylene plastic performed the best. It was a more flexible plastic which meant that it was easier to mold to the liner and it did not leave any residue that was unable to come off with a wipe.

Conclusions

This project is divided into two designs: concrete and cob-specific. Each meets the constraints set by our client, Hana Mori. The designs create reusable, accessible mold systems using adaptable liners, modular molds, and low-waste materials. We addressed the unique curing processes of each material while maintaining quality, reducing waste, and ensuring ease of reuse and user accessibility.

For the COB molds, the push-up mechanism, combined with removable and washable fabric liners, creates a simple, affordable, and consistent design that requires minimal maintenance and allows for the preservation of the materials. By testing different fabric combinations and ventilation patterns, we optimized the design and identified the best prototype with a felt lining and diagonal ventilation holes, which best preserve specimen consistency and geometry while achieving the fastest curing period.

For the concrete molds, we tested sustainable materials and additional liners that can be added to existing systems to improve demolding ease compared to single-use traditional molds. The modular 3D-printed mold provided an adaptable, watertight structure that could accommodate multiple liner materials while maintaining a simple, consistent assembly process. Through qualitative analysis of the different liners, polypropylene performed best overall, maintaining specimen quality with the simplest cleaning process across additional iterations. Additionally, the wood-supported clay formwork demonstrated further potential for low-waste fabrication by using natural materials to form and cure concrete.

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